USN | | | | | | | | | | [23CSBSPH03]

FIRST SEMESTER B. TECH EXAMINATION, JANUARY - 2025

FUNDAMENTALS OF PHYSICS

Time: 3 Hours **Maximum Marks: 100**

Instructions:

i. Missing data may be suitably assumed.

ii. Physical constants

(i) Mass of the electron $m_e = 9.1 \times 10^{-31} kg$

(ii) Mass of proton, neutron $m_n, m_p = 1.67 \times 10^{-27} kg$

(iii) Charge of the electron $e = 1.602 \times 10^{-19} C$

(iv) Planck's constant $h = 6.626 \times 10^{-34} Js$

(v) Permittivity of free space $\varepsilon_o = 8.85 \times 10^{-12} F/m$

(vi) Velocity of light $c = 3 \times 10^8 m/s$

(vii) Permeability of free space $\mu_o = 4\pi \times 10^{-7} h/m$

ANSWER ALL QUESTIONS

			Blooms Level	CO	Marks
		PART-A	**5 × 2=10**		
1.	**a.**	Describe linear periodic motion and circular periodic motion.	L-2	CO1	**02**
	b.	Paraphrase the diffraction of light.	L-2	CO2	**02**
	c.	Discuss Heisenberg's uncertainty principle.	L-2	CO3	**02**
	d.	Describe characteristic properties of lasers.	L-2	CO4	**02**
	e.	Define zeroth law of thermodynamics.	L-1	CO5	**02**
		PART-B	**6 × 5=30**		
2.	**a.**	Obtain an expression for force constant of two springs connected in parallel.	L-2	CO1	**05**
		OR			
	b.	Set up the differential equation for Simple harmonic oscillator.	L-2	CO1	**05**
3.	**a.**	Discuss plane, elliptical and circularly polarized light.	L-2	CO2	**05**
		OR			
	b.	Distinguish the interference and diffraction phenomenon of light.	L-2	CO2	**05**
4.	**a.**	Summarize atomic packing factor for SC, and FCC structures.	L-2	CO3	**05**
		OR			
	b.	State and explain de-Broglie hypothesis on matter waves.	L-2	CO3	**05**

Deemed-to-be University
Learning Resource Centre
Jain Global Campus

5.	a.	Describe the construction of Ruby laser with neat diagram.	L-2	CO4	05
		OR			
	b.	Explain the construction of optical fibers with neat diagram.	L-2	CO4	05
6.	a.	Evaluate the Planck's constant using light emitting diodes Given Grating constant c= $3 \times 10^8 m/s, e = 1.6 \times 10^{-19} C$.	L-4	CO4	05

Color of LED	Wavelength (λ m)	Knee voltage (V_k in Volts)
Blue	346×10^{-9}	3.5
Green	538×10^{-9}	2.27
Yellow	568×10^{-9}	2.08
Red	630×10^{-9}	2.05
IR	940×10^{-9}	1.37

		OR			
	b.	Discuss any two types of optical fiber with neat diagrams.	L-3	CO5	05
7.	a.	Discuss the change in entropy in reversible and irreversible processes.	L-2	CO5	05
		OR			
	b.	State and explain second law of thermodynamics	L-2	CO5	05
		PART-C	**6 × 10=60**		
8.	a.	Deduce an expression for the amplitude and phase of the forced oscillations.	L-2	CO1	10
		OR			
	b.	**i.** What are damped oscillations?	L-1	CO1	02
		ii. Obtain the equations for decay of the amplitude in damped oscillations.	L-3	CO1	08
9.	a.	**i.** Describe the theory of interference fringes using superposition principle.	L-3	CO2	07
		ii. The critical angle of incidence of water for total internal reflection is 48° for certain wavelength of light. What is the polarizing angle and the angle of refraction for light incident on water at an angle that gives maximum polarization of the reflected light?	L-3	CO2	03
		OR			
	b.	**i.** Differentiate Fresnel's and Fraunhofer's diffraction	L-2	CO2	04
		ii. Describe Fraunhofer diffraction at single slit. Arrive at the condition for principal maxima.	L-2	CO2	06

10.	**a.**	Assuming Schrödinger's time independent wave equation discuss the solutions for energy and Eigen functions of a particle in one dimensional potential well of width "a".	L-3	CO3	**10**
		OR			
	b.	Set up time independent one-dimensional Schrodinger's wave equation.	L-3	CO3	**10**
11.	**a.**	**i.** Describe the seven crystal systems with a neat diagram. **ii.** Define the concepts of conductors, semiconductors, and insulators according to the band theory of solids.	L-2 L-2	CO3 CO3	**07** **03**
		OR			
	b.	**i.** Obtain an expression for d-spacing in terms of miller indices **(h k l)** **ii.** The velocity of an electron was measured to be 9×10^5 m/s with an uncertainty of 2%. What is the uncertainty involved in the measurement of position.	L-3 L-3	CO3 CO3	**07** **03**
12.	**a.**	**i.** Derive an expression for numerical aperture of optical fibers with neat diagram. **ii.** Define population inversion and metastable state.	L-3 L-2	CO4 CO4	**07** **03**
		OR			
	b.	**i.** Derive an expression for Energy density using Einstein's coefficients. **ii.** The average output power of laser source emitting a laser beam of wavelength 7135 A° is 20mW. Find the number of photons emitted per second by the laser source.	L-3 L-3	CO4 CO4	**07** **03**
13.	**a.**	**i.** Describe the first law of thermodynamics. **ii.** Describe the Continuity equation for current densities	L-2 L-2	CO5 CO5	**05** **05**
		OR			
	b.	Derive Maxwell's equations in vacuum and non-conducting medium.	L-3	CO5	**10**